

Regulatory DNA Prediction from Peak and Context Sequences Using BiLSTMs and Transformers

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Abstract

Modern deep learning models achieve strong performance in predicting regulatory activity from genomic DNA sequences, but it remains unclear whether these predictions are driven by core regulatory elements or by surrounding genomic context. We study this question using DNase-seq peaks from the ENCODE project mapped to the GRCh38 reference genome.

We evaluate BiLSTM and Transformer architectures under three input conditions: peak-only sequences, context-only sequences, and full sequences combining both regions. This controlled experimental design allows us to isolate the contribution of core regulatory regions versus their surrounding genomic environment.

Across all input types, the BiLSTM consistently outperforms the Transformer. Peak-only inputs yield the highest performance (AUROC up to 0.945), followed by full sequences, while context-only inputs perform worst but remain above chance prediction. These results indicate that surrounding genomic context contains predictive signal even in the absence of core regulatory regions, highlighting the importance of carefully controlling for contextual information in genomic machine based learning.

1. Introduction

Regulatory DNA elements such as promoters and enhancers play a central role in gene regulation by controlling when and where genes are activated. These elements operate within complex genomic environments where both local sequence motifs and broader chromatin context influence regulatory activity. Regions of open chromatin are strongly associated with active regulatory elements and can be measured using assays such as DNase-seq, which profile chromatin accessibility across the genome and are widely used as labels in computational genomics tasks.

Modern advances in deep learning have enabled accurate prediction of regulatory activity directly from raw DNA sequences. However, the biological basis of these predictions remains difficult to interpret. In particular, it is unclear to what extent models rely on local regulatory information versus broader genomic regions.

This motivates the central question of this work: to what extent can regulatory DNA activity be predicted using only surrounding genomic context? Addressing this question is important for understanding whether modern models capture true regulatory signals or instead exploit contextual structure in genomic sequences.

To investigate this, we analyze DNase-seq peaks from the ENCODE project [1] mapped to the GRCh38 human reference genome [2].

We construct controlled input conditions that explicitly separate core regulatory regions from their surrounding genomic context, allowing for a comparison of their predictive contributions. Specifically, we evaluate three input settings: peak-only sequences, context-only sequences, and full sequences combining both regions. We further compare BiLSTM and Transformer architectures under each condition to assess how model design influences regulatory prediction based on genomic information.

2. Related Work

Recent studies have explored deep learning architectures for regulatory and functional genomic prediction. EpiBERT [3] introduces a multi-modal Transformer model that integrates DNA sequence and chromatin accessibility data to improve regulatory prediction across diverse cell types. This work highlights the importance of incorporating genomic context beyond the raw sequence alone.

Transformer-based models have also been widely used for predicting gene expression and other regulatory signals from DNA sequences. Pipoli et al. [4] demonstrate that Transformer architectures can capture long range dependencies in genomic sequences when trained on both sequence and regulatory features. These results suggest that attention-based models are well suited for modeling complex genomic interactions.

More broadly, large-scale genomic foundation models such as AlphaGenome [5] show that incorporating long genomic contexts improves predictive performance across multiple regulatory tasks, including chromatin accessibility and transcription factor binding prediction. However, these approaches also raise questions about interpretability, particularly regarding the exact biological information upon which models base their predictions off of.

Overall, prior work demonstrates strong progress in applying deep learning to regula-

tory genomics, while also motivating the need to better understand how much predictive performance depends on true regulatory sequences versus broader genomic context.

3. Methods

3.1. Dataset and Sequence Extraction

We constructed our dataset using DNase-seq narrowPeak annotations from the ENCODE project, which provide experimentally found regions of open chromatin across the human genome. These peaks were then mapped to the GRCh38 reference genome to ensure consistent genomic coordinates and standardize sequence extraction.

For each DNase-seq peak, we extract DNA sequences centered on the peak region. To study the role of genomic information, we define three distinct input types: peak-only sequences corresponding to the core regulatory region, context-only sequences consisting of flanking genomic regions excluding the peak, and full sequences that include both the peak and its surrounding context. This setup allows us to isolate the contribution of regulatory signal from its surrounding genomic environment.

All sequences are encoded using a simple integer mapping scheme, where nucleotides are mapped as A=1, C=2, G=3, T=4 before being passed into the models. Zeros were used for unknown nucleotides within DNA regions as well as sequence padding. This encoding scheme preserves sequence structure while allowing for efficient numerical representation for DNA sequences. Positive examples correspond to DNase-seq peaks, while negative examples were sampled from matched genomic background regions to maintain class balance. The final dataset consisted of 50,255 positive examples and 48,291 negative examples, which were divided into training and validation sets using an 80/20 split.

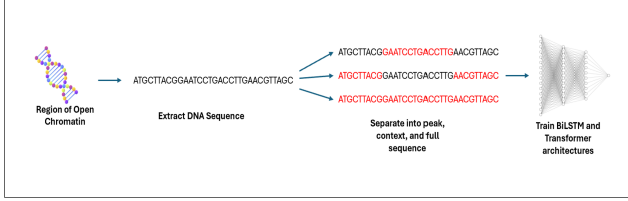


Figure 1. Overview of the experimental design. Regulatory DNA prediction was evaluated using peak-only, context-only, and full-sequence input conditions across BiLSTM and Transformer architectures.

3.2. Experimental Design

We design a controlled experimental framework to evaluate the role of genomic context in regulatory DNA prediction. Specifically, we compare three input conditions as described above: peak-only, context-only, and full-sequence inputs. This allows us to directly compare how much predictive signal comes from the main, high chromatin sequence and how much comes from the surrounding DNA.

We train and evaluate two architectures, a BiLSTM and a Transformer, under each input type. This results in six total experiments. All experiments use the same dataset splits, model capacities, and evaluation metrics to ensure fair comparison across all models and conditions.

This setup isolates the effect of inputted genomic information and model architecture, allowing us to assess whether genomic context alone can support regulatory prediction.

3.3. Model Architectures

We evaluate two sequence modeling architectures commonly used in genomic prediction tasks: a bidirectional long short-term memory (BiLSTM) network and a Transformer encoder.

The BiLSTM processes DNA sequences in both forward and backward directions, enabling it to capture local and mid-range dependencies within genomic sequences. The final hidden representations are passed through fully connected layers to produce a binary classification.

The Transformer model represents the more

standard genomic model architecture, and uses self-attention to capture global dependencies across the whole sequence. Positional encoding is applied to preserve spatial information, and the resulting learned representations are accumulated and passed through a classification head.

Both models receive identical input representations to ensure a fair comparison.

3.4. Training and Evaluation

All models were implemented using the PyTorch deep learning framework and trained using binary cross-entropy loss with the Adam optimizer. A fixed training and validation split was used to ensure consistent evaluation across experiments.

Model performance is evaluated primarily using AUROC (Area Under the Receiver Operating Characteristic curve), which measures the ability of a model to distinguish between regulatory and non-regulatory sequences across thresholds. Validation accuracy is also reported for completeness, but AUROC is treated as the primary metric due to class balance considerations.

Training dynamics are tracked across epochs to analyze convergence behavior and stability across different inputs and architectures.

4. Experiments and Results

4.1. BiLSTM Performance

Figure 2 summarizes the performance of the BiLSTM model across the three input conditions. The peak-sequence model achieved the strongest performance, reaching a validation accuracy of 0.870 and an AUROC of 0.945. The full-sequence model achieved an AUROC of 0.921, while the context-only one performed the worst with an AUROC of 0.770.

Across all evaluation metrics, the ordering of performance remained consistent, with peak-only sequences outperforming full sequences, and full sequences outperforming context-only sequences.

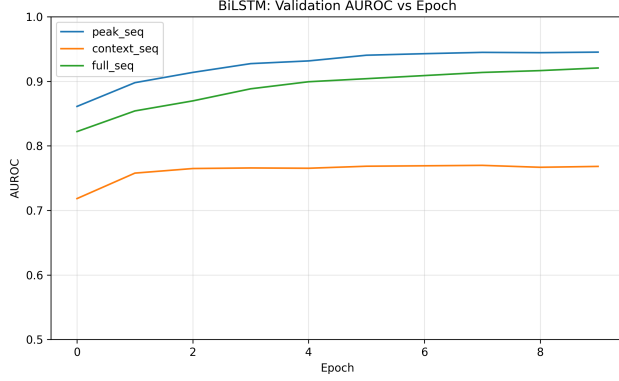


Figure 2. Validation AUROC across training epochs for BiLSTM models trained on different sequence inputs.

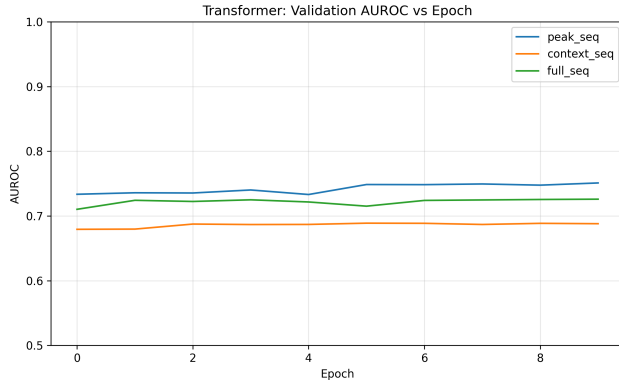


Figure 3. Validation AUROC across training epochs for Transformer models trained on different sequence inputs.

4.2. Transformer Performance

The Transformer model exhibited the same overall trend across input types. Peak-only sequences achieved the highest performance, with a validation accuracy of 0.677 and an AUROC of 0.751. The full-sequence condition achieved an AUROC of 0.726, while the context-only model achieved an AUROC of 0.689.

As with the BiLSTM, performance was highest under the peak-only input and lowest under the context-only input. Figure 3 shows the AUROC performance curves for all Transformer experiments.

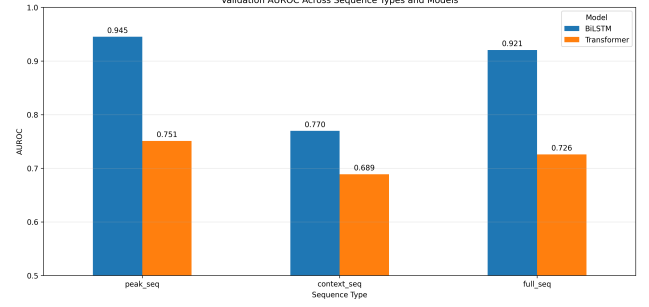


Figure 4. Comparison of model performance across all sequence input conditions.

4.3. Architecture Comparison

A comparison of both architectures across all input types is shown in Figure 4. The BiLSTM achieved higher validation accuracy and AUROC than the Transformer for peak-only, context-only, and full-sequence inputs.

The largest difference was observed in the peak-only condition, where the BiLSTM achieved an AUROC of 0.945 compare to 0.751 for the Transformer. Similar Performance gaps were observed for the full-sequence and context-only conditions. Despite these differences in performance, both architectures exhibited the same ordering of input conditions.

4.4. Summary of Results

The results in Figure 5 reveal two main patterns across all experiments. First, peak-only sequences produced the highest performance, followed by full sequences and then context-only sequences. Second, the BiLSTM outperformed the Transformer under every input condition and evaluation metric.

Although performance was reduced for context-only inputs, both architectures achieved AUROC values above 0.5, indicating predictive performance above random chance. Overall, the observed trends were consistent across models and input conditions.

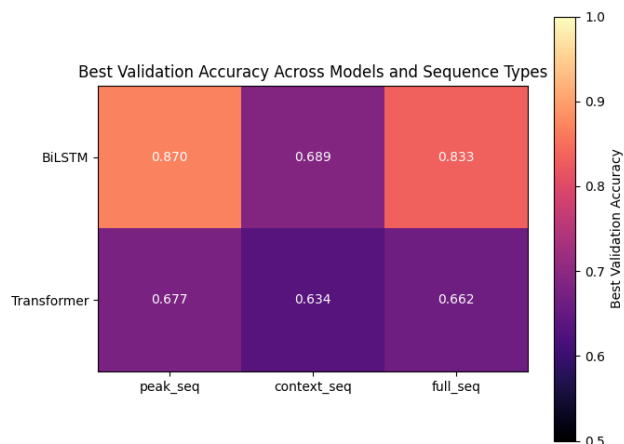


Figure 5. Heatmap of validation accuracy across models and sequence types.

5. Discussion

5.1. Role of Genomic Context

The primary goal of this project was to determine whether regulatory DNA activity can be predicted using only surrounding genomic context. While context-only models performed worse than models containing regulatory peaks, they consistently achieved performance above random chance. This indicates that surrounding genomic regions contain information associated with regulatory activity even when the core regulatory sequence is removed.

At the same time, peak-only sequences produced the strongest performance across both architectures. This suggests that the most informative signals remain concentrated within regulatory regions themselves. Together, these findings indicate that genomic context contributes useful predictive information, but that it is not as informative as the regulatory sequences identified by DNase-seq peaks.

5.2. Model Comparison

Across all experiments, the BiLSTM consistently outperformed the Transformer in both validation accuracy and AUROC. The largest performance difference occurred under the peak-only condition, where the BiLSTM achieved an AU-

ROC of 0.945 compared to 0.751 for the Transformer.

One possible explanation is that the dataset size and sequence lengths used in this project favored recurrent modeling approaches. BiLSTMs are effective at capturing local sequence patterns and medium range dependencies, while Transformers often benefit from larger datasets and pretraining. Despite differences in absolute performance, both architectures exhibited the same ordering of input conditions, suggesting that the observed trends were driven primarily by the information content of the sequences rather than model choice alone.

5.3. Limitations and Future Work

Several limitations should be considered when interpreting these results. First, this project used a single DNase-seq dataset and therefore reflects only one measure of regulatory activity. Second, the models were trained from scratch and do not represent the scale of modern genomic models. Finally, DNA sequences were represented using a simple integer encoding scheme, which may limit the ability of the models to learn richer sequence representations.

Future work could evaluate additional datasets, larger Transformer architectures, and pretrained genomic models to determine whether these findings generalize across tasks and scales. Interpretability methods could also be applied to identify which sequence features contribute most strongly to predictions under each input condition.

6. Conclusion

This project investigated the role of genomic context in regulatory DNA prediction using BiLSTM and Transformer architectures trained on DNase-seq peaks from the ENCODE project. By comparing peak-only, context-only, and full-sequence inputs, the contribution of regulatory regions and surrounding genomic context was evaluated separately.

The results showed that peak-only sequences

produced the strongest predictive performance, while context-only sequences remained above random chance, indicating that surrounding genomic regions contain meaningful predictive information. Additionally, the BiLSTM consistently outperformed the Transformer across all input conditions.

Overall, these findings suggest that regulatory prediction models rely primarily on signals contained within regulatory regions, but that genomic context also contributes to model performance. Understanding the relationship between regulatory sequences and contextual information remains important for improving the interpretability and biological relevance of deep learning models in genomics.

References

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